

Endless Life of Stone

An Embodied Journey into the Universe within the Void

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Abstract

Endless Life of Stone is an interactive installation blending classical Eastern philosophy with generative AI to challenge the perception of stones as static objects. Grounded in the cycle of "Formation, Existence, Destruction, and Emptiness," it illustrates a stone's endless ecological evolution—from dust aggregation and natural erosion to a lush cave ecosystem, before returning to nothingness. Technically, we achieved "digital weathering" via AI interpolation between hand-drawn keyframes. To translate the spatial philosophy of "the Universe within the Void (壺中天地)" into embodied perception, a computer vision-based distance mapping mechanism controls the observation scale. Viewing from afar reveals macro-geological evolution; stepping closer immerses viewers in the stone's micro-ecosystem; stepping away collapses the world to dust. This work invites viewers to physically measure the cycle of life, offering a new paradigm for digital humanities.

CCS Concepts

• **Human-centered computing** → **Interaction design theory, concepts and paradigms**; Visualization theory, concepts and paradigms.

Keywords

Interactive Art; Embodied Interaction; Generative AI; Eastern Philosophy; Digital Humanities

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1 Introduction

As the ancient proverb states, "A fist-sized stone can evoke a thousand miles of landscape (一拳之石，而能引万里之思)."

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Challenging the view of stones as lifeless, traditional Chinese philosophy recognizes them as active participants in ecological evolution [1]. Endless Life of Stone uses generative AI and interactive art to digitally explore this vitality, illustrating the life cycle of "non-living" matter.

The narrative maps the Four Kalpas—"Formation, Existence, Destruction, and Emptiness (成、住、坏、空)"—to a stone's life stages [2]. This traces its path from "Formation" (dust accumulation) and "Existence" (stable presence), through "Destruction" (erosion creating habitable cavities), to "Emptiness" (ultimate disintegration back to dust).

To visualize this, AI-generated transitions between hand-drawn keyframes simulate natural erosion. Beyond a visual effect, this evokes the "Universe within the Void," revealing a vibrant microcosm nurtured within the stone's crevices [3].

Blurring the boundaries between natural evolution and digital generation, we employ a distance-based embodied interaction mechanism. Physical proximity dictates the observation scale: viewing from afar reveals macro-geological formation, while stepping closer immerses viewers in the internal habitat. This transforms passive observers into active explorers, using bodily movement to navigate the "Universe within the Void" and experience life's endless cycle.

2 Philosophical Framework and Artistic Intent

2.1 The Cycle of Rebirth

In Buddhist cosmology, the "Four Kalpas"—Formation, Existence, Destruction, and Emptiness—posit that matter is never destroyed, but perpetually transforms [4]. Our installation maps this philosophical cycle onto a stone's ecological lifespan:

(1) Formation: The narrative begins with minute sand grains slowly compressing into solid entities, symbolizing the initial condensation of life's energy. **(2) Existence:** The stone enters a stable period. As the earth's skeleton, it provides fundamental physical support for the surrounding habitat. **(3) Destruction:** Erosion fractures the solid stone into porous cavities. Rather than demise, this mirrors classical Taihu stone aesthetics [5]. These voids become vital habitats, echoing "being-towards-death"—where decay engenders vitality. **(4) Emptiness:** Reaching its physical limit, the stone disintegrates back to dust. This is not nothingness, but potential energy nurturing the next cycle [6].

Visualizing millennia of evolution, this work illustrates that stones are not passive landscape elements, but vibrant, engaged living entities.

2.2 The Nested Spaces

Spatially, this work explores "the Universe within the Void", where tiny enclosed spaces house infinite universes. We visualize this nesting ecologically: micropores support microbes, mesopores guide airflow, and caves shelter complex life.

Breaking 2D constraints, our embodied interaction maps a viewer's physical approach to a journey from macro-geology to micro-ecology. This perspective shift embodies the dialectic of "the void and the tangible," revealing a vast, vibrant ecosystem concealed within the stone's solid shell, as shown in Fig.1.

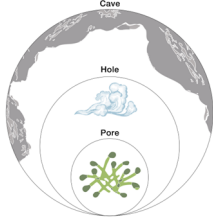


Figure 1: the Universe within the Void

3 System Design and Interactive Mechanism

3.1 Generative Pipeline

The visual assets for Endless Life of Stone rely on a deep human-AI symbiosis rather than random AI generation. Human artists establish the aesthetic and ecological logic, while generative AI enhances expressive power:

3.1.1 Human Intent. To accurately convey traditional Chinese cultural essence, we avoided pure text-to-video generation. Instead, we manually illustrated approximately 10 high-resolution keyframes to anchor the aesthetic framework, as shown in Fig.2.

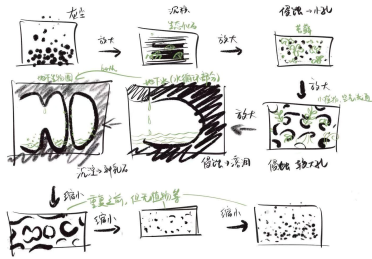


Figure 2: Sketches of hand-drawn keyframes

These keyframes strictly follow a visual evolution from microscopic dust and fossilized bones to solid rock formations, and finally to lush cave habitats. These hand-drawn illustrations infused the AI model with traditional Chinese aesthetic sensibilities and a specific spatial narrative, as shown in Fig.3.

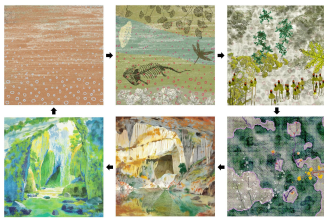


Figure 3: Hand-drawn keyframes

3.1.2 Digital Weathering. Using Vidu AI, we interpolate between anchored keyframes to create smooth transitions. This "digital weathering" compresses millions of years of erosion, dynamically computing material collapse and plant growth. Ultimately, this pipeline generates a seamless, looping visual stream. Embodying a continuous "dust to dust" ecological cycle, it serves as the fully navigable foundation for our distance-driven interaction.

3.2 Stepping into the Microcosm

To overcome the limitations of linear video playback, we employ an embodied interaction paradigm. A local computer vision pipeline (using an HD camera, Python, and MediaPipe) tracks the user's bounding box in real-time. This bounding box area estimates physical proximity, which is normalized and directly mapped to the AIGC video's frame index, enabling zero-latency navigation along the ecological timeline, as shown in Fig.4:

(1) Macro View (>2m): Displays a macro-geological loop of rock formation ("Formation" and "Existence") from a traditional subject-object distance; **(2) Crossing the Boundary (1–2m):** As viewers approach, proximity drives digital weathering, forming Taihu stone pores and blurring macro-micro boundaries; **(3) Stepping into the Void (<0.5m):** Reaching the threshold triggers a dramatic zoom, penetrating the stone's shell to envelop viewers in a lush cave—embodying the "Universe within the Void"; **(4) Ecological Collapse (Exit Reset):** Echoing "Emptiness," the unobserved ecosystem instantly dissolves to dust upon the viewer's departure.

Through this distance-mapping, physical movement replaces linear time. Viewers use their bodily steps to actively measure the birth and demise of matter.

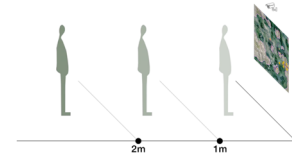


Figure 4: Distance-Driven Scale Mapping

4 Conclusion

Endless Life of Stone reconstructs a non-anthropocentric narrative via generative AI and embodied interaction, reimagining stones as active ecosystems. By mapping the Four Kalpas and Taihu stone aesthetics onto AI latent space, we visualize "digital weathering," breathing new life into ancient concepts. Furthermore, we overcome the limitations of 2D screens: our distance-driven mechanism uses the viewer's physical presence to navigate this digital ecosystem. Intertwining human intent with machine generation, the installation invites audiences to rethink human-nature connections, offering a new digital humanities paradigm that seamlessly blends Eastern heritage with technological ingenuity.

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